ChatGPT

BVTech.org Comprehensive Digital Audit and Overhaul Plan

Executive Summary

BVTech.org is already unusually ambitious for a small managed-services firm: it combines a commercial MSP website, transparent pricing, case studies, security tools, a cyber range, a gamified Academy, coding labs, certificates, community features, and location-specific service pages under one domain. The commercial homepage communicates meaningful differentiators—13+ years in business, published pricing, a documented implementation process, real project photography, reviews, and a 2023 SuperOps award—and the Academy has grown into a substantial learning product with six tracks, 129 stated lessons, 26 coding labs, 19 CTF challenges, exams, capstones, XP, BVPoints, certificates, pets, and leaderboards. [1]

The site's biggest problem is therefore **not lack of content**. It is that several distinct products have accumulated faster than the information architecture, design system, content governance, SEO taxonomy, educational model, and measurement framework have matured around them. Navigation currently mixes commercial buying journeys, tools, games, Academy functions, resources, certificates, community, and external portal experiences. The homepage alone exposes a large number of navigation choices, while Academy navigation differs between pages; the Labs page also contains duplicated or closely overlapping subjects such as two incident-response entries, multiple VPN/firewall concepts, overlapping AI-agent/RAG/prompting lessons, and similar introductory-versus-advanced topics without an explicit prerequisite system. [2]

The highest-value overhaul is to turn BVTech into **two tightly linked but clearly separated experiences**:

1. **BVTech Business** — managed IT, cybersecurity, cloud, networks, projects, pricing, locations, proof, and lead generation.
2. **BVTech Academy** — structured learning paths, 500 labs, assessments, credentials, cyber range, coding environments, and gamification.

They can remain on the same domain to retain authority and cross-linking benefits, but they should use visibly distinct navigation states, content models, conversion goals, analytics funnels, and page templates.

The recommended implementation is a staged **14–18 week program**, not a big-bang redesign. Weeks 1–3 establish inventory, analytics, URL mapping, design tokens, automated regression tests, and schemas. Weeks 4–7 rebuild shared navigation and the highest-value commercial templates without changing existing URLs. Weeks 6–11 introduce the new Academy taxonomy and learning-data model while retaining compatibility adapters for current XP, BVPoints, accounts, certificates, and labs. Weeks 10–14 migrate content and deploy the first expanded lab cohort. Weeks 14–18 cover localization, remaining redirects, monitoring, accessibility remediation, performance tuning, and gradual lab expansion.

The target should be **500 high-quality labs, not 500 thin pages**. Every lab should have explicit prerequisites, difficulty, estimated time, learning objectives, NICE-aligned competency metadata where relevant, an observable task, rubric, remediation path, evidence of completion, and version ownership. NIST's NICE Framework is expressly intended as a common language for cybersecurity work, education, training, skills, and workforce development, making it a strong backbone for BVTech's cyber curriculum. [3]

The highest-priority work is:

Priority	Task	Effort	Target
P0	Freeze URL deletion; build full URL/content/redirect inventory	Medium	Week 1
P0	Establish automated regression, Lighthouse, accessibility and link checks	Medium	Weeks 1–2
P0	Define Business vs Academy architecture and canonical URL rules	Medium	Weeks 1–2
P0	Instrument GA4/GTM events and funnel definitions	Medium	Weeks 1–3
P0	Build shared metadata/schema/canonical/breadcrumb component	Medium	Weeks 2–4
P1	Simplify global navigation and CTA hierarchy	Medium	Weeks 3–5
P1	Rebuild service, location, case-study and Academy templates	High	Weeks 4–8
P1	Normalize Academy taxonomy/prerequisites and eliminate duplicates	High	Weeks 5–9
P1	Implement lab data model and assessment engine	High	Weeks 6–11
P1	WCAG 2.2 AA remediation	High	Weeks 4–10
P1	Core Web Vitals/performance program	Medium	Weeks 4–10
P1	Security hardening and automated dependency/secret scanning	Medium	Weeks 3–8
P2	Spanish commercial localization with hreflang and parity rules	Medium	Weeks 8–12
P2	Publish first 100 new labs using production template	High	Weeks 10–14
P2	Add learning paths, badges, transcripts and LMS interfaces	High	Weeks 10–16
P3	Expand toward 500 validated labs in controlled cohorts	High	Weeks 14+
P3	Digital PR/backlink program around tools, research and Academy	Medium	Ongoing

The program should be considered successful only if it improves both business and educational outcomes: qualified assessment bookings, phone/contact conversions, location/service organic visibility, Academy registration-to-first-lab completion, path completion, retention, assessment integrity, accessibility, Core Web Vitals, and crawl/index health.

Detailed Audit

Content, information architecture and conversion

BVTech's strongest commercial content is concrete rather than generic. The homepage shows real project outcomes, a documented nine-stage delivery process, transparent service pricing, calculators, client proof, and a specific large venue project; the case-study and review surfaces reinforce this with named projects and customer feedback. [4] This is considerably more persuasive than standard MSP copy built around undifferentiated claims such as "trusted partner" and "tailored solutions."

The weakness is **content sprawl and hierarchy dilution**. The homepage navigation currently presents Services, Pricing, Case Studies, Reviews, a Resources dropdown with tools, Explore, security scan, calculator, email checker, pricing, compare, case studies, Trust Center, booking, guides and news, plus an Academy dropdown with games, answers, labs, Lightning Labs, coding labs, Cyber Range, challenges, leaderboards, community, pets and certificate verification. [5] Several choices duplicate destinations already in the top level or compete with the primary sales action.

There are also content-governance issues. The Academy's main page advertises "129 deep lessons," while its Labs search result has historically described "30+" labs, and the current Labs listing contains large track inventories. The Labs page itself contains overlapping titles—Incident Response appears twice, VPN concepts appear more than once, firewall topics overlap, and the AI track includes introductory and advanced versions of prompting, RAG, agents and LLM concepts without visible prerequisite relationships. [6] Those are signs that the Academy has outgrown a flat-content model.

Recommendation: establish one authoritative content registry for labs, lessons, coding exercises, CTFs, guides, news articles, services, locations, case studies and tools. Page counts, cards, navigation, structured data and sitemaps should be generated from that source rather than separately hard-coded.

SEO, GEO and AEO

The current search footprint has several strong foundations. BVTech already has dedicated service pages, city pages for San Antonio, Houston and Austin, a blog, guides, reviews, case studies, pricing and a founder entity page. [7] Google recommends understandable site structures, canonical URL consolidation, XML sitemaps and structured data where appropriate; BVTech has enough content depth to benefit substantially from disciplined implementation of those practices. [8]

The city pages currently contain useful service-area specificity, but some copy drifts into generic city exposition—for example broad statements about Houston's economy or San Antonio's population rather than unique BVTech operating evidence. [9] Location pages should earn their indexation through genuinely local proof: response model, project examples, industries served, neighborhoods/coverage, local testimonials, travel/on-site policy, locally relevant risk guidance and city-specific FAQs.

A Spanish homepage exists at `/es/`, but it is much shallower than the English ecosystem and even leaves part of the footer in English. [10] This is enough to demonstrate localization intent, but not yet a full bilingual architecture.

Backlink evidence visible from public search includes BVTech's LinkedIn company profile, a DesignRush profile and SuperOps' award site. [11] This audit does **not** have access to a commercial backlink index such as Ahrefs, Semrush or Majistic, so domain-rating, referring-domain counts and toxic-link conclusions would be speculative. A proper implementation phase should import Search Console links plus one independent commercial backlink dataset.

For AEO, BVTech has excellent source material—FAQs, definitions, calculators, lab explanations, guides and first-party operational expertise—but should make answers much more machine-extractable. Google supports structured-data features including Organization, LocalBusiness, Breadcrumb, Course list and other educational/search formats, while Schema.org defines `LearningResource`, `Course` and `EducationalOccupationalCredential`. [12] Structured data should describe visible truth, not be used as a ranking shortcut.

UX/UI, graphics, accessibility and mobile

The site has a distinctive personality. Real photography, project figures, gamification, icons, tools and interactive calculators make it memorable, while the Academy's pets, XP, BVPoints and cyber-range concepts create a stronger identity than a typical training catalog. [13]

The downside is competing visual emphasis. The primary business header can expose phone, email, language switching, search, booking, multiple top-level menu items and deep dropdown options simultaneously. [5] On Academy pages, additional game and platform choices amplify the density. A conversion-oriented visitor should not have to decide among a dozen equally emphasized actions.

Recommended desktop header:

Business mode:
`BVTech | Services | Solutions | Industries | Locations | Proof | Resources | Academy`
Right side: `Call` + primary **Book Assessment**

Academy mode:
`BVTech Academy | Learn | Practice | Paths | Cyber Range | Community | Credentials`
Right side: `Search` + `Sign in` + primary **Start Learning**

Mobile should collapse into task-first groups rather than reproduce a mega-menu verbatim.

Accessibility should target **WCAG 2.2 AA**, including keyboard operation, logical focus order, visible focus, accessible names, contrast, target sizes, error messages, reduced-motion handling and semantic structure. W3C's current WCAG 2.2 guidance explicitly covers keyboard accessibility and focus-order requirements. [14] High-risk BVTech components include dropdown navigation, before/after sliders, calculators, code editors, interactive game controls, modal dialogs, accordions, leaderboards and dynamically loaded labs.

Performance, security and analytics

The site appears to use Google Tag Manager; GTM is present in the rendered homepage and Academy pages. [15] Public rendering alone does not reveal whether GA4 events, consent behavior, attribution, cross-subdomain tracking or conversion definitions are configured correctly.

Performance should be governed by field-oriented Core Web Vitals. Google's current Core Web Vitals are LCP, INP and CLS, evaluated at the 75th percentile for passing field performance. [16] Adopt internal budgets tighter than the failure thresholds:

Metric	BVTech budget
LCP p75 mobile	≤2.2 s target, never >2.5 s
INP p75	≤180 ms target
CLS p75	≤0.08 target
TTFB cached	≤500 ms
Initial JS	≤170 KB compressed on content pages
Initial CSS	≤60 KB compressed
Hero image	≤180 KB AVIF/WebP
Noncritical third-party JS	lazy/consent-loaded
Total initial transfer	≤1.0 MB typical commercial page
Fonts	≤2 families, ≤4 critical files

Security review should use OWASP ASVS as the application-control baseline; OWASP describes ASVS as a requirements and verification framework for web application security, and its secure-header guidance covers defenses such as HSTS. [17] Because the Academy has authentication, code execution, assessments, certificates, gamification and community functions, it deserves a stricter application-security program than the public marketing pages.

Recommended controls include HSTS; CSP with nonces/hashes rather than broad unsafe directives; `frame-ancestors`; `X-Content-Type-Options`; restrictive `Referrer-Policy` and `Permissions-Policy`; secure/HttpOnly/SameSite cookies; CSRF protection; server-side authorization on every progression/reward operation; rate limiting; abuse controls; dependency and secret scanning; signed certificate verification; isolated code sandboxes; sanitized user content; immutable audit trails for rewards/exams; backup restoration testing; and separate production, staging and ephemeral-preview secrets.

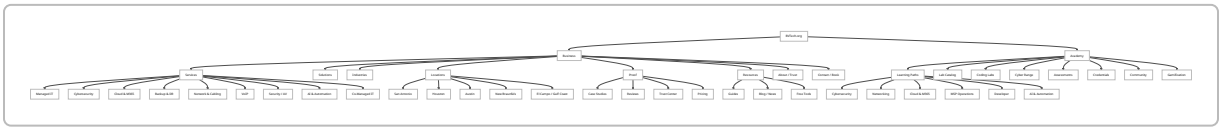
Key-page current-versus-recommended comparison

Page	Current state	Recommended state
`/`	Rich proof, pricing, calculator, services and multiple CTAs; very dense navigation. [5]	Hero → proof strip → outcomes → 3 core solution groups → case study → pricing anchor → process → reviews → FAQ → single primary assessment CTA.
`/services/`	Broad list of many offerings. [18]	Service hub grouped into Managed IT, Security, Cloud, Infrastructure, Communications, Projects; cards include audience, outcome and proof.
`/services/managed-it-services/`	Detailed inclusions, pricing context and FAQ. [19]	Add “best for,” pains/outcomes, deliverables, onboarding timeline, SLA model, proof, pricing range, comparison links, city applicability and conversion module.
`/pricing/`	Strong transparency and ranges. [20]	Keep URL; simplify packages, expose assumptions, calculator, example configurations, FAQ and assessment CTA.
`/san-antonio/`, `/houston/`, `/austin/`	Dedicated localized pages, but substantial generic city copy. [21]	Replace generic city facts with local project evidence, response model, local sectors, testimonials and city/service cross-links.
`/about/`	Detailed founder narrative and credentials. [22]	Condense biography; emphasize company governance, operating principles, team model, certifications, proof and trust links.
`/reviews/`	80+ claimed verified reviews and detailed testimonials. [23]	Add filters by service/industry/location; link each relevant review to service/case study; avoid unsupported self-serving review rich-result expectations.
`/academy/`	Strong gamified proposition with six learning tracks and multiple activity types. [24]	Make “choose goal → placement → learning path → lab → assessment → credential” the dominant experience.
`/academy/labs/`	Large flat track lists with duplicated/overlapping concepts. [25]	Searchable lab catalog with track, domain, level, duration, modality, prerequisite, role, technology and completion filters.
`/academy/code/`	26 browser coding labs with tests and rewards. [26]	Normalize into same lab object; include starter files, tests, hints, competencies, rubric and sandbox metadata.
`/cyber-range/`	Separate challenge/tier experience. [27]	Integrate as “Practice → Cyber Range”; map every challenge to skills, path prerequisites and credentials.

Page	Current state	Recommended state
`/es/`	Partial Spanish landing experience. [10]	Create parity for core commercial pages; implement locale selector, hreflang, translated metadata and human QA.

Target Architecture, SEO and Content System

Recommended site map



The key architectural decision is **not** to move existing high-value URLs merely for cosmetic consistency. Google treats permanent redirects as a canonicalization signal, but unnecessary migrations create crawl and ranking risk. [28] Keep established service and city URLs unless there is a strong semantic reason to change them.

URL and canonical rules

Use:

`/services/{service}/`
`/industries/{industry}/`
`/{city}/` for established city URLs; do not gratuitously move them to `/locations/`.
`/case-studies/{client-or-project}/`
`/guides/{topic}/`
`/blog/{slug}/`
`/tools/{tool}/`
`/academy/paths/{path}/`
`/academy/labs/{lab-slug}/`
`/academy/assessments/{assessment}/`
`/academy/credentials/{credential}/`

Rules:

- HTTPS only.
- One trailing-slash convention.
- Lowercase ASCII slugs.
- No dates in evergreen URLs.
- Self-referencing canonical on indexable pages.
- Query-filter combinations canonicalize to the unfiltered catalog unless intentionally indexable.
- Search/account/progress/game-state URLs should be `noindex` where appropriate.
- XML sitemap indexes split by commercial pages, resources, Academy, labs and images.
- Every redirect must be one hop; eliminate chains.

- Removed content with a real successor → 301.
- Truly discontinued content → 410 only after internal links are removed.
- Never redirect unrelated retired pages to the homepage.

Google recommends canonical URLs in sitemaps and treats canonical annotations and redirects as signals for duplicate consolidation. [29]

Metadata and schema

Homepage

Title: `Managed IT & Cybersecurity in Texas | BVTech LLC`
Description:
`Award-winning Texas managed IT, cybersecurity, cloud and network services for small and midsize businesses. Transparent pricing, 24/7 monitoring and founder-led support.`

Schema graph: `Organization` + appropriate `LocalBusiness` representation only for genuine physical business locations + `WebSite`. Google recommends Organization markup on the homepage or an organization/about page and advises providing real-world identity properties such as name, address where applicable, telephone, URL and logo. [30]

Service page

Title formula: `{Service} for Texas Businesses | BVTech LLC`
Schema: `Service`, provider `Organization`, `BreadcrumbList`, relevant `FAQPage` semantics only where accurately represented; do not assume FAQ markup will produce a visible rich result.

Location page

Title: `Managed IT & Cybersecurity in {City}, TX | BVTech`
Schema: organization/service-area data; use `LocalBusiness` only when that page represents a bona fide business location with a real address, not merely a service area. Google's LocalBusiness guidance requires a physical address for eligible local-business markup. [31]

Guide/article

`Article` / `BlogPosting`, `Person` author, `Organization` publisher, `datePublished`, `dateModified`, `headline`, representative image and breadcrumbs.

Academy path

`Course` + provider + learning outcomes + prerequisites + credential relationship. Google's Course structured-data guidance expects genuine courses and requires valid course names/providers for its supported presentation. [32]

Lab

Use a `LearningResource`-oriented schema model with visible properties for educational level, learning-resource type, competencies, time required, author/provider and language. Schema.org defines `LearningResource` specifically for works oriented toward learning, education and skill acquisition. [33]

Credential

`EducationalOccupationalCredential` with credential category, competency requirements, issuer and verification URL where semantically accurate. [34]

Internal linking

Every service page should link to:

- 2–3 related services.
- applicable location pages;
- one case study;
- pricing;
- one relevant guide/tool;
- assessment CTA.

Every location page should link to:

- the 4–6 highest-demand services locally;
- local or nearest applicable proof;
- reviews/case studies;
- pricing;
- contact.

Every guide should link up to:

- parent topic hub;
- relevant service;
- relevant free tool;
- 2–4 related guides;
- optional Academy learning path.

Every lab should link to:

- prerequisite lab(s);
- next lab;
- skill/domain hub;
- associated path;
- relevant glossary/reference;
- credential requirements.

Breadcrumbs should represent a meaningful user path rather than mechanically mirror the URL hierarchy, consistent with Google's guidance. [35]

Keyword clusters

Commercial clusters should be organized by intent rather than repeated city-keyword permutations:

Cluster	Primary intent	Supporting topics
Managed IT	Buy	managed IT services Texas, outsourced IT, IT support, co-managed IT
Cybersecurity	Buy/compare	SMB cybersecurity, MDR, endpoint security, phishing training, incident response
Microsoft 365	Buy/solve	M365 management, tenant security, Conditional Access, migrations
Backup/DR	Solve/buy	BCDR, immutable backup, business continuity, RPO/RTO
Network	Buy	network design, Wi-Fi, VLANs, structured cabling, Meraki, UniFi
VoIP	Buy	business phones, Dialpad, 3CX, AI receptionist
Vertical	Buy	healthcare IT, legal IT, construction IT, professional services IT
Local	Buy	managed IT San Antonio/Houston/Austin etc.
Academy	Learn	cybersecurity labs, networking labs, cloud labs, MSP training
Career	Learn	SOC skills, help desk skills, cloud skills, MSP technician path
Tools	Use	subnet calculator, phishing checker, security assessment
Expertise	Research	NIST, zero trust, M365 security, ransomware, backup testing

Avoid creating hundreds of near-duplicate city/service doorway pages. Expand location content only where BVTech can add unique operational, project or market evidence.

AEO content format

Each important informational page should begin with a compact "answer block" directly under the H1:

> **What is managed IT?** Managed IT is an ongoing service in which a provider monitors, maintains, secures and supports a business's technology environment for a recurring fee. BVTech's model includes monitoring, patching, endpoint security, help desk, vendor management and technology planning.

Then provide:

- "In short" 40–70-word answer.
- key facts table.

- “Who this is for.”
- process.
- cost/range when known.
- common alternatives.
- FAQ.
- cited primary sources for standards-based topics.
- visible author/reviewer.
- updated date.

This improves human comprehension and gives search and answer systems clean, attributable passages without writing robotic “SEO copy.”

Localization

The existing Spanish page proves demand is anticipated but lacks parity. [10] Adopt `/es/` as the Spanish root and maintain mirrored URLs for the top commercial pages first:

`/es/servicios/`
`/es/servicios/ti-administrada/`
`/es/ciberseguridad/`
`/es/san-antonio/`
`/es/houston/`
`/es/austin/`
`/es/precios/`
`/es/contacto/`

Use reciprocal `hreflang="en-US"` and `hreflang="es-US"` plus `x-default`; translate titles, descriptions, schema text, form validation and CTAs; do not auto-redirect based on IP/browser language; preserve a persistent language switcher; and use human review for technical terminology.

Academy and 500-Lab Expansion

The Academy already has the ingredients of a strong learning system: content tracks, hands-on code, CTFs, exams, capstones, certificates and a persistent reward system. [36] The next stage is to convert this from a collection of activities into a **competency graph**.

Proposed taxonomy

Domain	Target labs	Example subdomains
Cybersecurity Foundations & Blue Team	85	IAM, malware, SIEM, threat hunting, IR, DFIR, governance
Network Engineering	70	TCP/IP, switching, routing, DNS, Wi-Fi, VPN, firewalls, SD-WAN

Domain	Target labs	Example subdomains
Cloud & Microsoft 365	75	AWS, Azure, M365, Entra, containers, Kubernetes, IaC, FinOps
MSP Operations	60	RMM, PSA, onboarding, ticketing, BCDR, vCIO, documentation
Software & Automation	65	Python, JS, APIs, SQL, Git, testing, CI/CD, automation
AI Engineering & Security	55	prompting, RAG, agents, evaluations, MLOps, AI security
Offensive Security / Cyber Range	45	recon, web, privilege escalation, exploitation, CTF scenarios
GRC, Privacy & Compliance	25	risk, policy, HIPAA concepts, PCI concepts, vendor risk
IT Support & Endpoint Operations	20	troubleshooting, Windows, identity, device management
Total	**500**	

Cybersecurity lab metadata should map, where appropriate, to NICE work roles, competency areas and task/knowledge/skill concepts. NIST explicitly positions NICE as a common language for cybersecurity workforce, career, training and education use. [37]

Learning paths

Create role-oriented paths on top of the domain taxonomy:

IT Support Technician → Systems Administrator → MSP Technician → Senior MSP Engineer

Networking Foundations → Network Technician → Network Engineer → Secure Network Architect

Cyber Foundations → SOC Analyst I → SOC Analyst II → Incident Responder → Threat Hunter

Cloud Foundations → Cloud Administrator → Cloud Security Engineer → Cloud Architect

Developer Foundations → Automation Engineer → DevSecOps Engineer

AI Foundations → AI Automation Builder → RAG Engineer → AI Security Practitioner

A learner may take individual labs freely, but credentials require a defined graph of prerequisite competencies.

Lab object

Each lab should be stored as structured content, for example:

```yaml
id: CYB-PHI-201
slug: analyze-business-email-compromise
title: Analyze a Business Email Compromise
version: 1.2.0
domain: cybersecurity
subdomain: phishing
level: intermediate
estimated_minutes: 35

prerequisites:
  - CYB-PHI-101
  - IAM-MFA-101

objectives:
  - Identify sender-domain anomalies
  - Distinguish display-name spoofing from domain spoofing
  - Recommend containment actions

competencies:
  nice:
    - "mapped-competency-id"

modality: investigation
environment: browser-sandbox

assessment:
  type: evidence_plus_quiz
  pass_score: 80
  attempts: 3

rewards:
  xp: 180
  bvpoints: 150

badge_progress:
  - phishing-analyst

author: bvtech
review_cycle_days: 180
```

Standard lab template

Every lab page:

1. Scenario and job context.
2. Objectives.
3. Prerequisites.
4. Estimated time and difficulty.
5. Environment/safety notice.
6. Task briefing.
7. Workspace.
8. Evidence submission.
9. Automated and/or rubric assessment.
10. Feedback.
11. Remediation/hints.
12. “Why this matters in the real world.”
13. References.
14. Next recommended activity.

Do **not** award substantive XP merely for loading or scrolling a lesson. The existing Academy already states that XP is intended to represent demonstrated achievement; preserve that principle. [24]

Assessment portfolio

Use multiple types rather than universal multiple choice:

Assessment	Best use
MCQ / multi-select	terminology and conceptual discrimination
Ordering	incident-response, troubleshooting and deployment processes
Matching	ports/protocols, control categories, service models
Short constructed response	reasoning and remediation
Configuration task	cloud/network/admin skills
Code tests	programming and automation
Artifact submission	diagrams, rules, queries, scripts
Log investigation	SOC/DFIR
Flag capture	cyber range
Scenario branching	support, governance, social engineering
Capstone	integrated job-role competence
Oral/reflection optional	communication and judgment

Scoring:

`final score = 35% task correctness + 25% evidence quality + 20% reasoning + 10% efficiency + 10% safety/procedure`

Not every lab needs every dimension. Knowledge quizzes can remain binary or percentage-based; complex labs should use explicit rubrics.

Suggested mastery bands:

- 90–100: Mastery
- 80–89: Proficient
- 70–79: Developing; remediation recommended
- <70: Not yet demonstrated

Credential exams should use larger randomized item banks, server-side grading and tamper-resistant attempt records.

Gamification

Keep XP, BVPoints, leaderboards, pets and Bitopia—the existing game identity is an asset. [24] Add guardrails so gamification rewards learning rather than grinding:

- XP comes primarily from assessed mastery.
- Repeated easy labs have sharply diminishing rewards.
- Daily streaks reward consistency, not hours online.
- Separate “knowledge,” “practice,” “community” and “challenge” achievements.
- Leaderboards offer opt-out/privacy controls.
- Seasonal events never gate core credentials.
- “Comeback” mechanics prevent streak loss from punishing adult learners.
- Badges require demonstrable competencies, not simple page visits.

Badge families:

Foundation, **Practitioner**, **Specialist**, **Professional**, **Capstone**, **Community**, **Challenge**, and **Career Path**.

Each serious skill badge should expose issuer, criteria, evidence/verification URL, issue date and version.

LMS integration

Use an adapter architecture:

```
BVTech Learning Event Bus
        |
        +-- Native Academy profile
        +-- xAPI/LRS adapter
```

```
        +-- LTI 1.3 adapter
        +-- Open Badges adapter
        +-- CSV/API enterprise reporting
```

1EdTech maintains LTI 1.3 as the current LTI family for tool/platform interoperability. [38] An LTI adapter would let institutions launch BVTech activities from a compatible LMS while preserving BVTech's own learning engine. xAPI-style event export can provide granular learning-event telemetry, while badges/certificates can remain independently verifiable.

Implementation order:

1. Internal normalized event schema.
2. REST/webhook enterprise reporting.
3. xAPI-compatible export.
4. LTI 1.3 launch and grade passback.
5. portable credential/badge integration.

Do not couple the Academy's core database directly to an LMS vendor.

Implementation Blueprint, Testing and Rollout

Page templates

Commercial service template

```
Breadcrumb
H1 + 40–70 word answer
Primary CTA / phone
Trust strip
Problems we solve
Outcomes
What's included
How onboarding works
Technology / standards
Relevant case study
Pricing range
Related services
Local coverage
FAQ
Final CTA
```

Location template

```
Breadcrumb
H1 + service-area statement
```

```
Local response/coverage facts
Top services
Local industries
Local proof
Testimonials
How onsite/remote support works
Pricing
Local FAQs
Related guides
Assessment CTA
```

Guide template

```
H1
Answer summary
Author / reviewer / updated
Key takeaways
Table of contents
Concept explanation
Decision/framework table
Implementation steps
Pitfalls
Primary references
Related tool
Related service
Related Academy path
```

Lab template

As specified in the Academy section, generated from structured lab metadata.

Image and visual specifications

Use real project photography wherever possible rather than generic stock.

- Master originals: ≥2400 px long edge.
- Content derivatives: AVIF first, WebP fallback.
- Hero: roughly 1600×900; ≤180 KB target.
- Card: 800×600 or 800×450; ≤90 KB.
- Logo assets: SVG.
- Icons: SVG sprite/component library rather than emoji as essential interface labels.
- Explicit `width` / `height` or `aspect-ratio` on every visual to prevent layout shift.
- Responsive `srcset`.
- Lazy-load below-fold imagery.
- Meaningful alt text for informative images; empty alt for decorative assets.

- Academy diagrams should have downloadable/text-equivalent descriptions.
- OG/social: 1200×630.
- Structured-data business imagery should include suitable high-resolution aspect ratios where applicable; Google's LocalBusiness guidance recommends multiple high-resolution image shapes including 1:1, 4:3 and 16:9. [31]

Analytics design

Keep GTM if operationally useful, but define a governed event dictionary.

Commercial:

```
service_view
pricing_view
calculator_started
calculator_completed
phone_click
email_click
assessment_cta_click
booking_started
booking_completed
contact_started
contact_submitted
case_study_view
guide_download
```

Academy:

```
academy_signup_started
academy_signup_completed
path_started
lab_started
lab_completed
lab_failed
hint_used
assessment_started
assessment_passed
assessment_failed
badge_earned
certificate_earned
return_7d
return_30d
```

Required dimensions: content type, service, city, path, lab ID, difficulty, acquisition source, device, authenticated/anonymous state without exposing personally identifying information into analytics.

Primary business funnel:

`Landing → Service/Pricing → Proof → CTA → Booking → Qualified lead`

Primary Academy funnel:

`Academy landing → Signup → Path selected → First lab started → First lab passed → 3 labs passed → Path milestone → Credential`

Testing stack

Recommended CI sequence:

```
npm ci
npm run lint
npm run typecheck
npm test
npm run build

npm run test:unit
npm run test:integration
npm run test:e2e
npm run test:a11y
npm run test:links
npm run test:schema

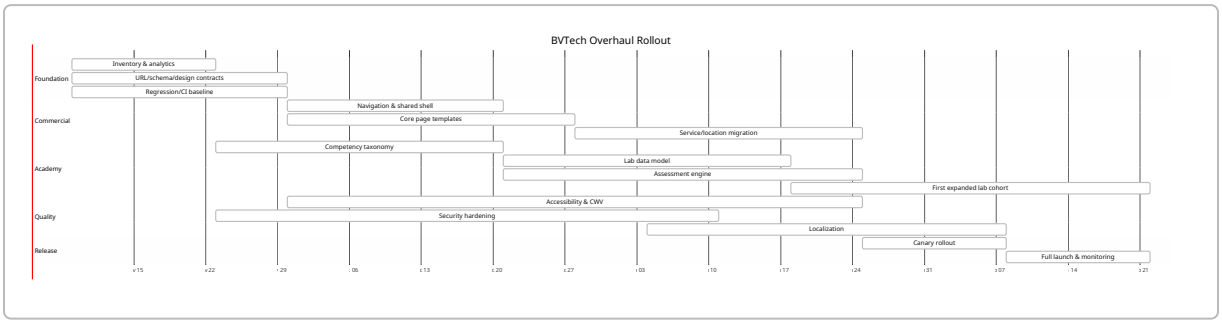
npx playwright test
npx lighthouse-ci autorun
```

Add:

- axe-core automated accessibility checks.
- Playwright keyboard-navigation tests.
- schema JSON-LD validation tests.
- broken-link crawler.
- redirect-map tests.
- canonical/hreflang tests.
- sitemap validation.
- visual-regression snapshots.
- mobile viewport suite.
- authentication/authorization tests.
- reward/idempotency tests.
- assessment cheating/tampering tests.
- code-sandbox isolation tests.
- dependency/secret scanning.
- synthetic monitoring post-deployment.

Automated accessibility testing is not sufficient; perform manual keyboard and screen-reader checks on at least the global header, menus, booking/contact flow, Academy signup, lab interaction, code editor, assessment, certificate verification and community controls. WCAG 2.2 remains the governing target. [39]

Rollout



The dates are illustrative scheduling anchors starting the first full week after this report; actual sequencing should follow repository discovery.

Rollout rules:

- No URL deletion before redirect inventory approval.
- Deploy shared shell behind a feature flag.
- Release templates to a small page subset first.
- Preserve current server endpoints and account APIs through adapters.
- Shadow-write learning events before migrating progress.
- Backfill progress in staging and compare counts.
- Canary Academy changes to staff/test accounts.
- Preserve certificate verification URLs permanently.
- Export database snapshots before every schema migration.
- Maintain rollback artifact for each release.
- Monitor 404s, 5xx, conversion, sign-in failures, reward issuance and Search Console coverage daily during launch.
- Do not publish hundreds of labs to search until each passes editorial/assessment QA.

Risk register

Risk	Impact	Mitigation
Organic traffic loss from URL changes	High	Keep URLs whenever possible; explicit 301 map; sitemap/canonical QA
Existing Academy progress lost	Critical	Immutable backups, migration dry runs, reconciliation scripts
XP/BVPoints exploitation	High	Server-authoritative rewards, idempotent transactions, audit logs

Risk	Impact	Mitigation
Certificate links break	High	Permanent IDs and legacy verifier routes
500 labs become thin/duplicative	High	competency graph, editorial gate, templates, version owners
Navigation redesign hurts leads	Medium	staged A/B or canary deployment; event comparison
Accessibility regression in games/ code editors	High	manual AT testing and component-level standards
Performance degrades as Academy grows	High	budgets in CI, route-level bundles, lazy loading
Spanish pages diverge from English	Medium	translation parity dashboard and owner
Schema becomes misleading	Medium	generate from visible page data; automated validation
Security regression in sandbox/ community	Critical	isolation, authorization testing, CSP, rate limits, ASVS verification
Analytics double fires during transition	Medium	event contract, debug mode, duplicate-event tests

Claude Code Implementation Prompt

Use the following prompt as the operating specification for Claude Code. It intentionally instructs the coding agent to discover the repository before changing frameworks or paths, because the public site does not reveal enough to safely assume the internal stack.

```
You are the senior engineer and technical migration lead for BVTech.org.

MISSION
Perform a production-safe overhaul of BVTech.org while preserving all working
functionality, existing indexed URLs, user accounts, Academy progress, XP,
BVPoints, badges, certificates, cyber-range functionality, coding sandboxes,
forms, calculators, analytics and integrations.

Do not perform a blind redesign. First inspect the repository, architecture,
routing, data stores, deployment configuration, tests and analytics.

PRIMARY OUTCOMES

1. Separate the UX into two coherent modes while retaining one domain:
   A. BVTech Business
   B. BVTech Academy
```

```
2. Simplify navigation:
   BUSINESS:
   Services | Solutions | Industries | Locations | Proof | Resources | Academy
   Right actions: Call + Book Assessment

   ACADEMY:
   Learn | Practice | Paths | Cyber Range | Community | Credentials
   Right actions: Search + Sign In + Start Learning

3. Preserve established URLs unless there is a compelling reason to migrate
them.

4. Establish a structured Academy content model capable of 500+ labs.

5. Target WCAG 2.2 AA.

6. Enforce performance budgets:
   - LCP target <= 2.2 s p75 mobile
   - INP target <= 180 ms
   - CLS target <= 0.08
   - initial JS <= 170 KB compressed on normal content pages
   - initial CSS <= 60 KB compressed
   - typical initial transfer <= 1 MB
   Do not weaken functionality just to game Lighthouse.

7. Harden security using OWASP ASVS principles.

8. Add complete automated testing, analytics event contracts and rollback
support.

PHASE 0 — REPOSITORY DISCOVERY

Before modifying code:

- Print directory tree excluding dependency/build directories.
- Identify:
  - framework/runtime and exact versions
  - router
  - templating system
  - CSS/design system
  - package manager
  - database(s)
  - authentication
  - Academy progress/reward storage
  - certificate implementation
  - code execution/sandbox implementation
  - API routes
  - deployment platform
```

```
  - environment-variable usage
  - analytics/GTM
  - sitemap/robots generation
  - schema/metadata components
  - tests and CI/CD
  - localization implementation

Run existing lint, test and build commands before editing.

Create:
docs/audit/repository-baseline.md

Record:
- current commands
- current failures
- current route list
- data migrations required
- high-risk modules
- rollback mechanism

Do not “fix” unrelated failing tests without recording them.

PHASE 1 — URL AND ROUTE INVENTORY

Generate machine-readable inventories:

docs/migration/routes-before.csv
docs/migration/redirect-map.csv
docs/migration/canonical-map.csv

For every public route store:
old_url
new_url
status
canonical
indexable
template_type
locale
notes

Rules:
- Keep existing high-value URLs whenever possible.
- Permanent move => 301.
- Do not use redirect chains.
- Do not redirect unrelated removed pages to /.
- Preserve certificate verification routes permanently.
- Preserve Academy legacy routes with aliases/adapters where needed.
```

```
Write an automated test that fails if a known legacy URL unexpectedly returns
404 or redirects more than once.

PHASE 2 — DESIGN TOKENS AND SHARED SHELL

Build/refactor a token system for:
- typography
- spacing
- radii
- borders
- shadows
- semantic colors
- focus states
- breakpoints
- motion

Do not scatter hard-coded style constants through pages.

Create accessible primitives:
Header
MegaMenu
MobileNav
Breadcrumb
Button
LinkButton
Card
Callout
Accordion
Tabs
Modal/Dialog
FormField
Alert
Badge
Pagination
FilterPanel

All interactive primitives must:
- work with keyboard
- expose proper accessible name/role/state
- show visible focus
- support reduced motion
- avoid focus traps except intentional modal focus management

Add skip navigation.

PHASE 3 — BUSINESS INFORMATION ARCHITECTURE

Implement templates:
```

```
ServicePage
LocationPage
IndustryPage
CaseStudyPage
GuidePage
ArticlePage
ToolPage
PricingPage

Suggested ServicePage order:
Breadcrumb
Hero + 40–70 word answer
Primary CTA
Proof strip
Problems
Outcomes
Included services
Process
Technology/security details
Case study
Pricing
Related services
Locations
FAQs
Final CTA

Suggested LocationPage:
Breadcrumb
Localized hero
Actual coverage/response information
Top services
Industries
Local proof
Testimonials
On-site/remote model
Pricing
FAQs
Related resources
CTA

Do not insert fabricated addresses, testimonials, metrics or service claims.

PHASE 4 — SEO / METADATA

Create a central metadata API/helper.

Each indexable route must emit:
```

```
- unique title
- unique description
- canonical
- robots directive
- OpenGraph metadata
- Twitter/social metadata if applicable
- structured data generated from page data

Implement:
Organization/WebSite schema on appropriate organization pages.
BreadcrumbList on hierarchical pages.
Service schema on services where semantically appropriate.
Article/BlogPosting on editorial content.
Course metadata on Academy learning paths.
LearningResource metadata on labs.
EducationalOccupationalCredential metadata on credentials.

Structured data must match visible content.

Generate XML sitemap indexes:
sitemap-pages.xml
sitemap-services.xml
sitemap-locations.xml
sitemap-resources.xml
sitemap-academy.xml
sitemap-labs.xml
sitemap-images.xml if useful

Add automated schema parsing/validation tests.

PHASE 5 — CANONICALIZATION / FILTERS

For Academy catalog filters:
- keep human-friendly query parameters
- canonicalize filter combinations to the base catalog unless explicitly
approved
  for indexing
- do not generate millions of crawlable filter combinations
- ensure pagination is crawlable
- search pages should normally be noindex

Use one lowercase trailing-slash convention based on the existing production
contract; do not change it unnecessarily.

PHASE 6 — ACADEMY DATA MODEL

Create a normalized Lab model similar to:
```

```
id
slug
title
version
status
domain
subdomain
difficulty
estimatedMinutes
summary
scenario
objectives[]
prerequisiteIds[]
competencies[]
learningPathIds[]
modality
environment
instructions
starterAssets[]
assessment
hints[]
references[]
reward
badgeProgress[]
author
reviewer
publishedAt
updatedAt
reviewDueAt

Do not copy this blindly if the existing architecture already has a good
content model. Adapt without data loss.

Lab IDs must be immutable.

Add migration/adapters so existing labs continue to work.

Normalize obvious duplicate concepts into:
- one canonical lab where identical, OR
- explicit foundational/advanced variants with prerequisites where pedagogically
  distinct.

Never delete learner completion records during normalization.

PHASE 7 — 500-LAB TAXONOMY

Seed taxonomy support for:
```

```
Cybersecurity Foundations & Blue Team: 85
Network Engineering: 70
Cloud & Microsoft 365: 75
MSP Operations: 60
Software & Automation: 65
AI Engineering & Security: 55
Offensive Security/Cyber Range: 45
GRC/Privacy/Compliance: 25
IT Support/Endpoint: 20

Total = 500

Do NOT generate 500 shallow pages automatically and publish them.

Instead:
- create the schema
- create curriculum manifest files
- create quality validation
- migrate existing labs
- produce the first production-quality expansion cohort
- make bulk content expansion repeatable

Every lab requires:
objectives
prerequisites
difficulty
duration
observable task
assessment
feedback/remediation
references
version owner

PHASE 8 — LEARNING PATHS

Implement graph-based learning paths.

Initial role paths:
IT Support Technician
Systems Administrator
MSP Technician
Network Engineer
SOC Analyst I
SOC Analyst II
Incident Responder
Cloud Administrator
Cloud Security Engineer
Cloud Architect
```

```
Automation Engineer
DevSecOps Engineer
AI Automation Builder
RAG Engineer
AI Security Practitioner

Provide:
- prerequisite graph
- progress percentage
- recommended next activity
- skill coverage
- credential requirements

Prevent cycles in prerequisite graphs with an automated validation command.

PHASE 9 — ASSESSMENTS

Support assessment types:
multiple_choice
multi_select
ordering
matching
short_response
configuration
code_tests
artifact
log_investigation
ctf
branching_scenario
capstone

All scoring that affects XP, credentials or leaderboard standing must be
server-authoritative.

Requirements:
- randomized pools where relevant
- attempt limits
- idempotent reward issuance
- audit trail
- anti-replay
- no reward from client-only state
- rubric support
- remediation messages
- accessible error/result output

Test duplicate submissions and concurrent completion requests.

PHASE 10 — GAMIFICATION
```

```
Preserve:
XP
BVPoints
leaderboards
BVPets
Bitopia
badges
certificates

Rules:
- XP should represent assessed achievement.
- repeated easy activities must not enable unlimited farming.
- seasonal rewards cannot gate core credentials.
- leaderboards need privacy/opt-out handling.
- streaks should have adult-friendly recovery mechanics.

Create transaction-style reward records:
reward_id
user_id
activity_id
reason
xp_delta
currency_delta
created_at
idempotency_key

PHASE 11 — LMS / LEARNING EVENTS

Create a normalized internal learning-event bus.

Events:
registered
path_started
lab_started
hint_used
lab_submitted
lab_passed
lab_failed
assessment_started
assessment_passed
assessment_failed
badge_earned
credential_earned

Do not tightly couple core learning logic to an LMS.

Create interfaces/adapters for:
```

```
- REST/webhook reporting
- xAPI-compatible statements
- LTI 1.3 integration
- portable badge/credential export

Stub adapters behind feature flags if credentials/configuration are unavailable.

PHASE 12 — ACCESSIBILITY

Target WCAG 2.2 AA.

Automated:
axe tests
semantic HTML tests
focus regression tests where practical

Manual-test checklist must cover:
- desktop nav
- mobile nav
- language selector
- forms
- calculators
- accordion FAQ
- before/after media
- Academy signup
- filters
- lab runner
- code editor
- assessment
- certificate verifier
- game controls
- community controls

Ensure:
- labels/instructions
- accessible validation
- keyboard navigation
- visible focus
- sufficient contrast
- target sizing
- reduced motion
- live regions for async feedback
- meaningful headings
- no color-only state

PHASE 13 — PERFORMANCE

Add Lighthouse CI or equivalent.
```

```
Implement:
- responsive images
- AVIF/WebP
- image dimensions/aspect-ratio
- lazy load noncritical media
- preload only true LCP assets
- font subsetting/preload discipline
- route-level code splitting
- lazy load games/editors/maps
- defer/consent-load noncritical third parties
- avoid hydration/JS on static content where architecture permits
- cache immutable assets
- preconnect only when measurable

Add CI budgets and fail on severe regressions.

PHASE 14 — SECURITY

Review against OWASP ASVS.

Implement/verify:
HTTPS-only
HSTS
CSP
frame-ancestors
X-Content-Type-Options
Referrer-Policy
Permissions-Policy
secure cookies
SameSite
CSRF defenses
input validation
output encoding
authorization checks
rate limits
login throttling
session rotation
secure password reset
secret scanning
dependency scanning
audit logging

Academy-specific:
- isolate code sandbox from application/network secrets
- resource/time limits on execution
- sanitize rendered output
- server-authoritative challenge checks
```

```
- prevent client-side certificate issuance
- rate-limit flag attempts
- protect progress/reward APIs against IDOR
- test privilege boundaries
- sanitize community content

Never weaken existing security controls for compatibility without documenting
the decision.

PHASE 15 — ANALYTICS

Keep existing analytics identifiers/configuration unless explicitly migrated.

Implement a typed/event-contract layer.

Business:
service_view
pricing_view
calculator_started
calculator_completed
phone_click
email_click
assessment_cta_click
booking_started
booking_completed
contact_started
contact_submitted
case_study_view
guide_download

Academy:
academy_signup_started
academy_signup_completed
path_started
lab_started
lab_completed
lab_failed
hint_used
assessment_started
assessment_passed
assessment_failed
badge_earned
certificate_earned

Do not send secrets, assessment answers or unnecessary PII to analytics.

Test that events fire once.
```

```
PHASE 16 — LOCALIZATION

Preserve existing English and Spanish routes.

Build locale infrastructure:
en-US
es-US

Use:
- reciprocal hreflang
- x-default
- localized title/descriptions
- localized validation strings
- persistent locale selector

Never use IP-based forced redirects.

Create a script/report that identifies untranslated production strings.

PHASE 17 — CI/CD

Adapt commands to the actual stack discovered in Phase 0.

Equivalent desired pipeline:

install
lint
typecheck
unit
integration
build
e2e
accessibility
link check
schema check
route/redirect check
performance smoke test
security/dependency scan

For a Node repository, likely commands resemble:

npm ci
npm run lint
npm run typecheck
npm test
npm run build
npx playwright test
npx lighthouse-ci autorun
```

```
Do not invent package scripts without adding them to package.json.

PR checks must block:
- build failures
- broken critical routes
- failing migrations
- redirect loops
- duplicate canonicals caused by regressions
- serious accessibility violations
- severe performance-budget regressions
- known high/critical dependency issues where remediation is available

PHASE 18 — DEPLOYMENT SAFETY

Before deployment:
[ ] database backup verified
[ ] migration dry-run passed
[ ] route diff reviewed
[ ] redirect map reviewed
[ ] sitemap generated
[ ] canonical tests passed
[ ] schema tests passed
[ ] accessibility smoke test passed
[ ] auth/progress/reward tests passed
[ ] certificate verifier tested
[ ] forms tested
[ ] analytics debug verified
[ ] performance budgets passed
[ ] security headers verified
[ ] rollback artifact exists

Deployment:
- use staging first
- use feature flags for navigation/Academy changes
- canary where architecture supports it
- run smoke suite against production immediately after release

Post-deployment:
[ ] monitor 404/5xx
[ ] monitor auth failures
[ ] monitor form submissions
[ ] monitor booking completion
[ ] monitor Academy completion events
[ ] reconcile reward issuance
[ ] validate Search Console indexing/sitemaps
[ ] inspect CWV/RUM
[ ] inspect logs/security alerts
```

```
ROLLBACK TRIGGERS

Rollback or disable feature flags if:
- elevated 5xx
- login/progress failures
- duplicate XP/currency
- certificate verification failures
- substantial lead-form failure
- redirect/canonical catastrophe
- sandbox isolation issue
- serious accessibility regression in a core path

DELIVERABLES

At completion produce:

docs/audit/repository-baseline.md
docs/architecture/site-map.md
docs/architecture/academy-model.md
docs/analytics/event-dictionary.md
docs/accessibility/manual-checklist.md
docs/security/asvs-review.md
docs/performance/budgets.md
docs/migration/routes-before.csv
docs/migration/redirect-map.csv
docs/migration/canonical-map.csv
docs/deployment/checklist.md
docs/deployment/rollback.md

Also provide:
- summary of files changed
- migrations applied
- commands executed
- test results
- remaining risks
- follow-up backlog

CRITICAL RULES

1. Preserve functionality before improving aesthetics.
2. Never silently delete existing routes.
3. Never invent business claims.
4. Never fabricate addresses or location schema.
5. Never migrate learner progress without reconciliation.
6. Never award meaningful credentials from client-only checks.
7. Never publish bulk AI-generated labs without editorial/assessment QA.
8. Prefer small, reviewable commits.
```

```
9. Keep compatibility adapters until production telemetry proves they are safe
to remove.
10. If repository reality conflicts with this prompt, preserve production
safety,
    document the conflict and implement the closest safe architecture.
```

Printable Stakeholder One-Page Summary

BVTech.org: From Feature-Rich Website to Unified Business + Learning Platform

What BVTech already has. BVTech combines a credible Texas MSP presence with unusually strong digital assets: transparent pricing, real project proof, 80+ stated reviews, free tools, guides, a Trust Center, dedicated city pages, a cyber range and a gamified Academy. The Academy already advertises 129 lessons, 26 coding labs, 19 CTF challenges, exams, capstones, XP, BVPoints, certificates, pets and leaderboards. [40] The commercial side also has tangible differentiation through real builds, service depth, transparent pricing and the independently hosted SuperOps 2023 MSP awards program. [41]

The core issue. BVTech has grown faster than its architecture. Business services, free tools, resources, games, learning content, community features and credentials compete inside dense navigation. Academy subjects overlap, localization is partial, and the platform needs a stronger shared model for metadata, analytics, accessibility, performance, security and learning progression. [42]

The new structure. Keep one authoritative domain but present two clear experiences: **BVTech Business** for buyers and **BVTech Academy** for learners. Preserve established URLs and existing accounts/progress. Business navigation becomes Services, Solutions, Industries, Locations, Proof, Resources and Academy. Academy navigation becomes Learn, Practice, Paths, Cyber Range, Community and Credentials.

The Academy opportunity. Expand to **500 assessed labs** across cybersecurity, networking, cloud/M365, MSP operations, software/automation, AI, offensive security, GRC and endpoint support. Labs become structured learning objects with prerequisites, objectives, duration, competency mapping, hands-on evidence, assessment, remediation and version ownership. Cybersecurity curriculum should use the NIST NICE Framework as a workforce-alignment vocabulary. [43]

Quality bar. Target WCAG 2.2 AA, strong Core Web Vitals, OWASP ASVS-aligned application security, server-authoritative Academy rewards and assessments, permanent certificate verification, structured metadata, disciplined canonicalization and automated regression testing. W3C defines WCAG 2.2 as the current accessibility recommendation set used here, Google defines LCP/INP/CLS as Core Web Vitals, and OWASP ASVS provides a verification framework for web application controls. [44]

Rollout. Use a **14–18 week staged implementation**, not a rewrite launch: establish inventory and tests first; introduce the shared design/SEO foundation; migrate commercial templates; normalize Academy data; ship the new learning/assessment architecture; add the first large lab cohort; then expand localization and the remaining lab catalog. Existing URLs, lead generation, user progress, XP/BVPoints, certificate URLs and working integrations remain protected by compatibility layers and rollback procedures.

Success measures. Commercial success means more qualified bookings, stronger organic service/location traffic and higher CTA completion. Academy success means a higher signup-to-first-lab rate, first-lab pass rate, path progression, 7/30-day retention, credential completion and demonstrable skill mastery. Platform success means fewer crawl errors, passing accessibility checks, improved Core Web Vitals, fewer broken links, zero reward duplication, safe code execution and predictable releases.

Strategic outcome: BVTech should stop behaving like a collection of good individual pages and features and start operating as a coherent platform: **a trust-rich MSP acquisition engine on one side and a competency-driven IT/cyber learning system on the other.**

---

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